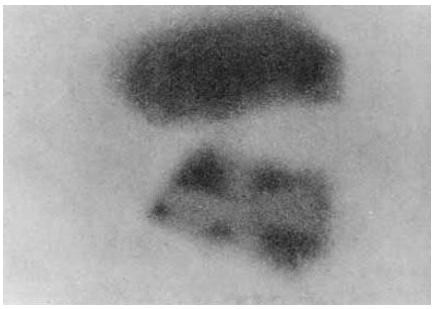
# Radioactivity

On February 26, 1896, a cloudy, dark day in Paris Becquerel could not carry out his experiment. The uranium compound he planned to use was placed on a photographic plate covered with black paper was in a drawer, and the key was in between the sample and the plate.



In 1896, Becquerel accidentally discovered that uranium salt crystals emit an invisible radiation that can darken a photographic plate even if the plate is covered and kept in dark.

His Ph.D. students Marie Curie & Pierre Curie carried out more research and discovered of two new radioactive elements **polonium** and **radium**. They received the 1903 Nobel prize in Physics for their research on radioactivity.

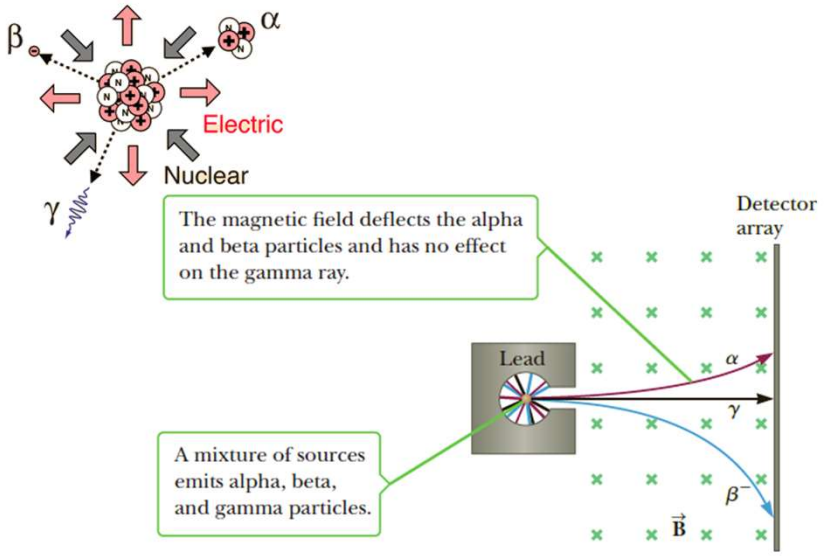


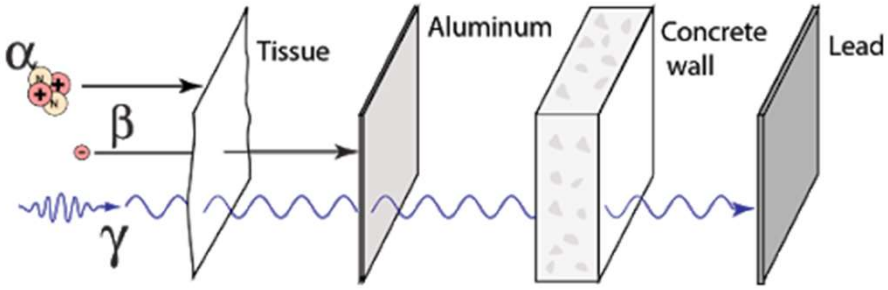
**Henri Becquerel**
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Nobel Prize 1/2

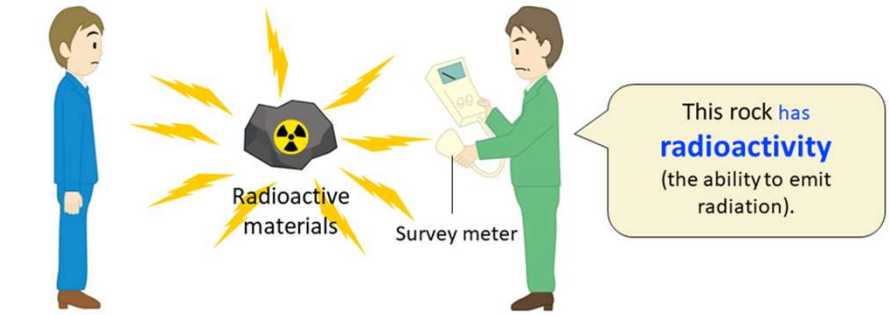
**Marie Curie**
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Nobel Prize 1/4

**Pierre Curie**
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Nobel Prize 1/4

**Radioactivity** is the result of the spontaneous decay (or disintegration) of unstable nuclei.

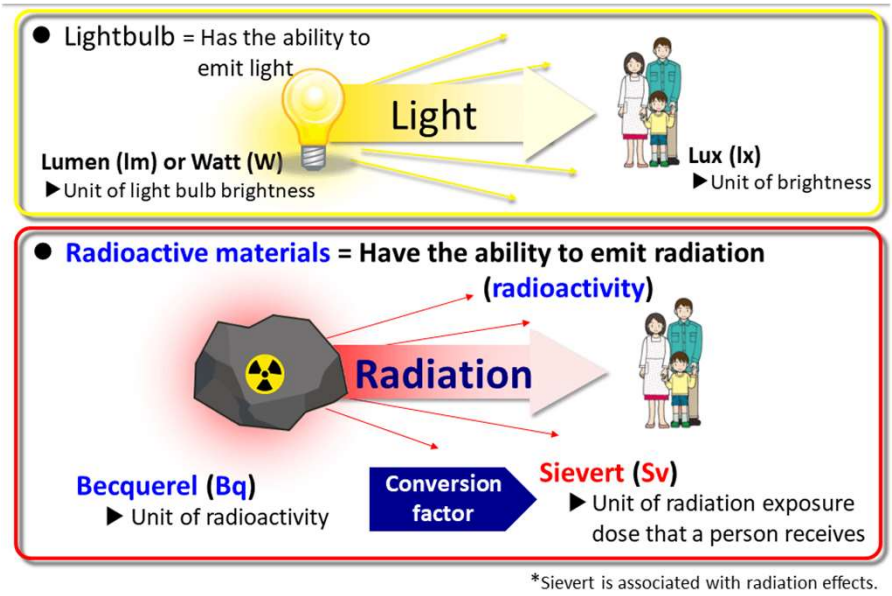





**Becquerel (Bq)**
**Unit for intensity of radiation:**
**one nucleus decays (disintegrates) per second = 1 becquerel**

**Sievert (Sv)**
**Unit of radiation exposure dose which a person receives:**
**associated with radiation effects**



## The radioactive decay law

The rate of decay (or disintegration) of a radioactive substance is proportional to the number of radioactive nuclei present in the sample.

$$-\frac{dN}{dt} \propto N$$

$$\frac{dN}{dt} = -\lambda N$$

integrate to obtain $N = N_o\, e^{-\lambda t}$

where $N_0$ is the initial number of nuclei,
$N$ is the number of radioactive nuclei at time $t$,
$\lambda$ is the decay constant (unit is $s^{-1}$)

## Activity (*A*)

The rate of decay or activity of a sample is defined as the number of decays per second.

$$A = \left|\frac{dN}{dt}\right|$$

$$A = \lambda N$$

If the the initial activity at time = 0 is $A_o$ and the activity at time = $t$ is $A$, then

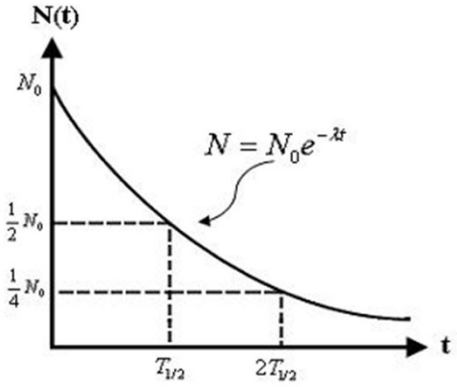
$$A_o = \lambda N_o$$

$$A = \lambda N$$

Therefore, $$\frac{A_o}{A} = \frac{N_o}{N}$$

## Half-life ($T_{1/2}$)

Half-life is the time required for radioactive nuclei in a sample to reduce to half of its initial value through exponential decay.




Obtain an expression for expression for $T_{1/2}$.

The decay constant is inversely related to the half-life ($T½$) by the formula

$$T_{1/2} = \frac{\ln 2}{\lambda}$$

$$T_{1/2} = \frac{0.693}{\lambda}$$

## Units of Radioactivity

The original (old) unit of activity is the curie (Ci)

1 Ci = $3.7 \times 10^{10}$ decays per second

The SI unit of activity is the becquerel (Bq)

1 Bq = 1 decay $s^{-1}$

## Decay Process

When one element changes into another by itself, the process is called the **spontaneous decay.**

As a general rule,

(1) The sum of the mass numbers A must be the same on both sides of the equation.

(2) The sum of the atomic numbers Z must be the same on both sides of the equation.

## Alpha Decay

- For alpha emission, the mass of the parent must be greater than the combined mass of the daughter nucleus and the alpha particle.
- In the decay process, this excess mass is converted into energy of other forms and appears as **kinetic energy** in the daughter nucleus and the alpha particle.
- Most of the kinetic energy is carried away by the alpha particle because it is much less massive than the daughter nucleus.

If a nucleus emits an alpha particle ($^4_2\mathrm{He}$), it loses two protons and two neutrons.

The atomic number Z decreases by 2 and becomes (Z-2)

The mass number A decreases by 4 and becomes (A-4)

$$^{A}_{Z}\mathrm{X} \rightarrow {}^{A-4}_{Z-2}\mathrm{Y} + {}^{4}_{2}\mathrm{He}$$

In such a decay X is called the **parent nucleus** and Y is called the **daughter nucleus**.

If Y is also radioactive, then the new nucleus is called **grand-daughter** nucleus.

$$^{238}_{92}\mathrm{U} \rightarrow {}^{234}_{90}\mathrm{Th} + {}^{4}_{2}\mathrm{He}$$

## e.g.: Decay of radium

$$^{226}_{88}\mathrm{Ra} \rightarrow {}^{222}_{86}\mathrm{Rn} + {}^{4}_{2}\mathrm{He}$$

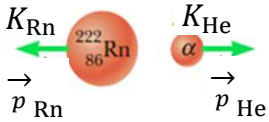
Before $\alpha$-decay




Kinetic energy $K_{\mathrm{Ra}} = 0$

Linear Momentum $\vec{p}_{\mathrm{Ra}} = 0$

After $\alpha$-decay




**Momentum is conserved** in these two particles and therefore, they move in **the opposite directions**.

The velocity of lighter particle is higher and has more kinetic energy than that of the massive particle.

## Beta Decay

In beta decay an unstable atomic nucleus transforms to a more stable one by emitting a beta particle [an electron ($\beta^-$) or a positron ($\beta^+$) ] and an antineutrino ($\bar{\nu}$) or a neutrino ($\nu$).

However, we know that a nucleus does not have $e^-$ or $e^+$.
There are two primary processes making beta decay.

(1) In **beta-minus decay**, a neutron transforms into a proton, emitting **an electron** and **an antineutrino**.

(2) In **beta-plus decay**, a proton transforms into a neutron, emitting **a positron** and **a neutrino**.

**In beta decay,**
the atomic number **Z** of the element **changes by** 1, resulting in a new element but the mass number **A remains the same**.

## Beta-minus Decay

- Occurs in nuclei with too many neutrons.
- A neutron changes into a proton.
- An electron ($\beta^-$ particle) and an electron antineutrino ($\bar{\nu}$) are emitted.

$$n \rightarrow p^+ + e^- + \bar{\nu}$$

- The atomic number Z increases by one (Z+1), and the atom changes to a different atom.

e.g.: Carbon-14 decays to nitrogen-14 via beta minus decay (neutron to proton),

$$^{14}_{6}C \rightarrow {}^{14}_{7}N^+ + \beta^- + \bar{\nu}$$

## Beta-plus Decay

- Occurs in nuclei with too many protons.
- A proton changes into a neutron.
- A positron ($\beta^+$ particle, the antimatter of an electron) and an electron neutrino are emitted.

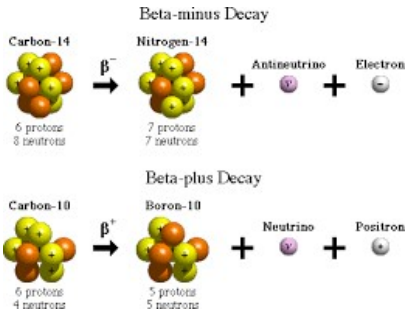
$$p^+ \rightarrow n + e^+ + \nu$$

- The atomic number **Z** decreases by one **(Z-1)**, and the atom changes to a different atom.

e.g.: Sodium-22 decays to neon-22 through beta plus decay (proton to neutron).

$$^{22}_{11}\mathrm{Na} \rightarrow {}^{22}_{10}\mathrm{Ne}^+ + e^+ + \nu$$

## Characteristics of beta decay




**(1) Emission of Beta Particles:**
Electrons ($\beta^-$) or positrons ($\beta^+$) are ejected from the nucleus at high speeds.

**(2) Conservation of Charge and Lepton Number:**
The total charge and lepton number are constant before and after the decay.

*..... continued*

**(3) Formation of Neutrinos/Antineutrinos:**
For the conservation of energy and momentum a neutrino or antineutrino is also emitted.

**(4) Change in Atomic Number:**
The number of protons changes, thus changing the element.

**(5) Constant Mass Number:**
Since the total number of nucleons remains unchanged, the mass number of the atom is the same.

**(6) Weak Nuclear Force:**
Beta decay is governed by the weak interaction (one of a fundamental forces of nature) .

## Gamma decay

- Gamma decay is the process where an excited atomic nucleus loses excess energy by emitting a gamma ray (a high-energy photon).
- It usually happens after alpha ($\alpha$) or beta ($\beta$) decay, which leaves the nucleus in an excited, unstable state.
- During gamma decay, the nucleus's mass number and atomic number remain unchanged and only releases energy.

$$^{12}_{5}\mathrm{B} \rightarrow {}^{12}_{6}\mathrm{C}^{*} + e^{-} + \bar{\nu}$$

$$^{12}_{6}\mathrm{C}^{*} \rightarrow {}^{12}_{6}\mathrm{C} + \gamma$$

where the asteric (*) in $^{12}_{6}\mathrm{C}^{*}$ indicates an excited nucleus.

# Applications of Radioactivity

If $A$ is the present activity and $A_o$ was the initial (past) activity, then using the equations

$$N = N_o\, e^{-\lambda t} \text{ and } A = \lambda N$$

we can obtain $A = A_o\, e^{-\lambda t}$

or $\frac{A_o}{A} = e^{\lambda t}$

$$\ln\left(\frac{A_o}{A}\right) = \lambda t$$

$$t = \frac{\ln\left(\frac{A_o}{A}\right)}{\lambda}$$

By measuring the ratio of two activities, the time difference between them can be estimated.

## Radioactivity has many diverse applications in following fields:

- archaeology and geology
- medicine
- industry
- agriculture and food

## (1) Carbon Dating

Carbon-14 dating is a technique that estimates the age of organic materials by measuring the amount of the radioactive carbon-14 ($^{14}C$) remaining in a sample.

**i. Formation of $^{14}C$:**

High-energy cosmic rays strike the atmospheric nitrogen atoms, creating radioactive $^{14}C$.

This $^{14}C$ reacts with oxygen to form carbon dioxide, which is absorbed by plants through photosynthesis.

Animals then intake this $^{14}C$ by eating plants.

**ii. Constant Ratio During Life:**

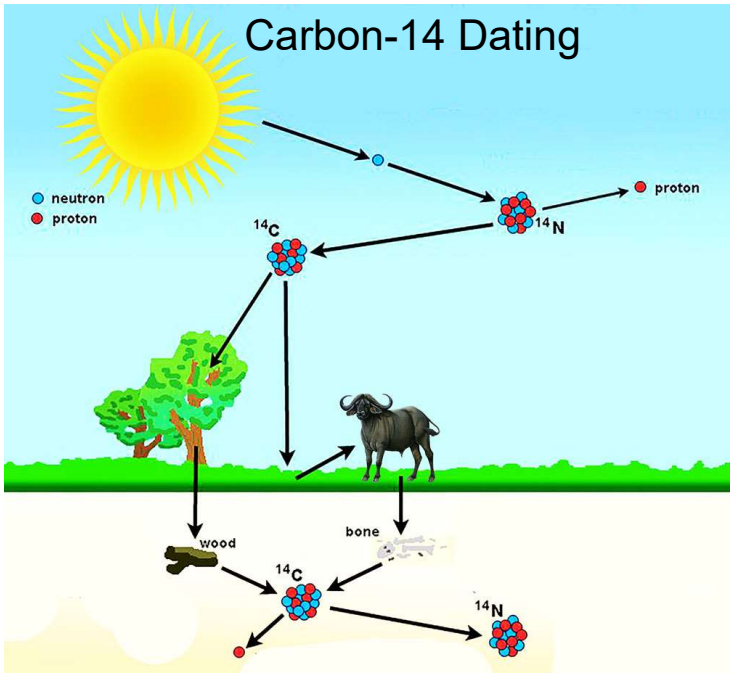
In living bodies, the ratio of $^{14}C$ to $^{12}C$ remains relatively constant and equal to the atmospheric $^{14}C$ : $^{12}C$ ratio which is 1 to 1 trillion ($1:10^{12}$).

**iii. Decay After Death:**

Once the body dies, the $^{14}C$ in its body begins to decay into nitrogen-14, with a half-life of 5,730 years.

**iv. Age Calculation:**

Scientists measure the remaining $^{14}C$ in an ancient sample and compare it with the $^{14}C$ in living sample to determine how many half-lives have passed.




## Applications of C-14 dating

- **Archaeology:**
  Carbon-14 dating is important for finding timelines of ancient civilizations by dating organic materials like wood, charcoal, and bone.

- **Geology:**
  The method helps find age of fossils, sediments, and other ancient organic remains to understand Earth's history and past climates.

## Limitations

- **Age Range:**
  Due to limited $^{14}C$ half-life, it can only be used to measure up to about 50,000 years old.
- **Contamination:**
  If the sample is affected or contamination with radiation it must be corrected.

# (2) Nuclear radiation in medicine

## (a) Diagnostic Applications (imaging)

i. **Nuclear Imaging:**

Radiopharmaceuticals are introduced into the body by mouth or by injection, to create detailed images of organ function and structure.

ii. **PET scans** (Positron Emission Tomography):

These scans visualize physiological processes, such as glucose consumption in the brain, to diagnose diseases like Parkinson's.

iii. **Use of Radioisotopes:**

Technetium-99m (Tc) is widely used for imaging the skeleton, heart, and other organs.

Iodine-131 is used for thyroid imaging and treatment.

## (b) Radiotherapy Applications

i. **Radiation therapy** uses intense energy to weaken or destroy cancerous cells, shrinking tumors.

ii. **External or Internal Radiation:**

External Beam Radiotherapy uses machines to aim high-powered beams of radiation at the tumor.

In Internal Radiotherapy, capsules of radioactive isotopes are directly introduced into or near the tumor to give a concentrated dose of radiation.

## (c) Sterilization of medical equipment.

Beams of gamma rays are used to kill bacteria and other pathogens for sterilizing surgical instruments and other medical equipment.

# Next Lecture

## (3) Applications in agriculture and food

**i. Identification of fertilizer usage:**
Use nitrogen-15 ($^{15}N$) isotopes to track the distribution of applied fertilizers in agricultural systems.

**ii. Genetic modification:**
Use ionizing radiation, like gamma rays, to introduce mutations in plant DNA to make genetically modified (GM) improved food verities.

**iii. Insect pest management**
Insects are sterilized using ionizing radiation and released back into pest-infested areas. The technique reduces reproduction and destroys insect pests.

**iv. Food safety and control**
Appling ionizing radiation to food kills harmful microbes and prevents the spread of insect pests.

## Radioactive Series

Natural radioactive series are chains of radioactive decays.

- **Parent Nuclide:** The initial radioactive isotope at the start of the series.
- **Daughter Nuclide:** A radioactive nuclide produced from the decay of a parent nuclide.
- **Stable Isotope:** An isotope that does not decay further, marking the end of the series.
- **Alpha Decay:** A type of radioactive decay that involves the emission of a helium nucleus, which reduces the mass number of the nuclide by 4.
- **Beta Decay:** A type of radioactive decay involving the emission of a beta particle, which can change the atomic number but not the mass number.

There are four radioactive series but only three are considered as natural series.
A parent nuclide transforms through a series of unstable daughter nuclides until it reaches a stable end product (lead isotopes for natural series).

1. **The Thorium Series (4n series):**
   Starts with natural $^{232}Th$ and ends with the stable lead isotope $^{208}Pb$.

2. **The Uranium Series (4n+2 series):**
   Starts with naturally occurring $^{238}U$ and ends with the stable $^{206}Pb$.

3. **The Actinium Series (4n+3 series):**
   Starts with the isotope $^{235}U$ and Ends with the stable $^{207}Pb$.

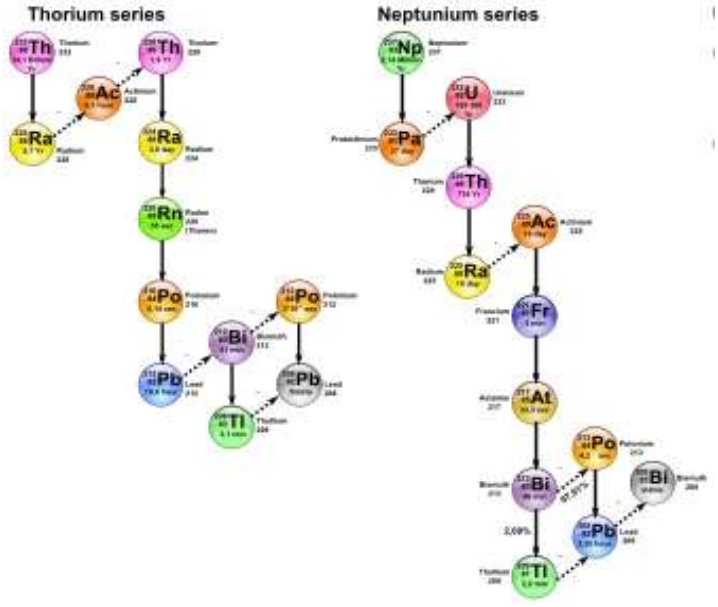
## The artificial Radioactive Series

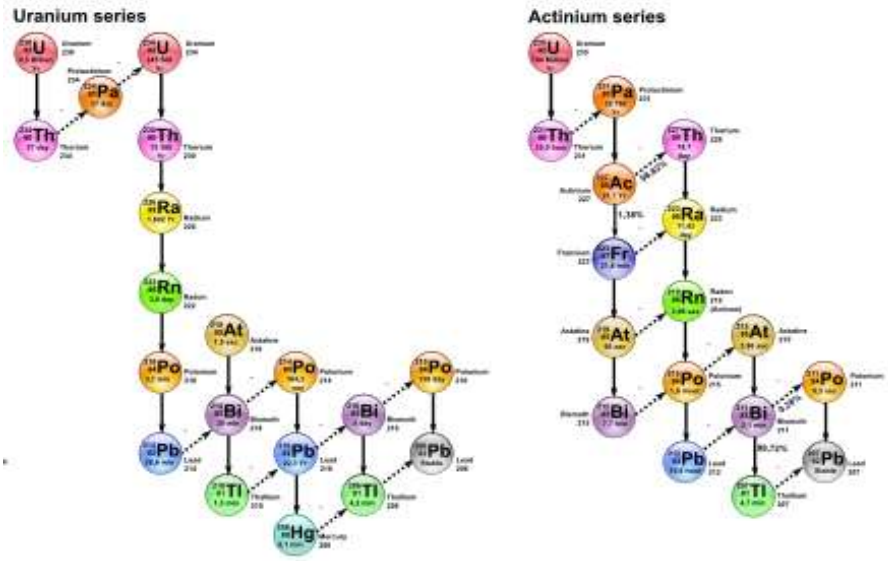
**4. The Neptunium Series (4n+1 series)**

Starts with $^{237}Np$ and ends with the isotope (Bismuth-209).

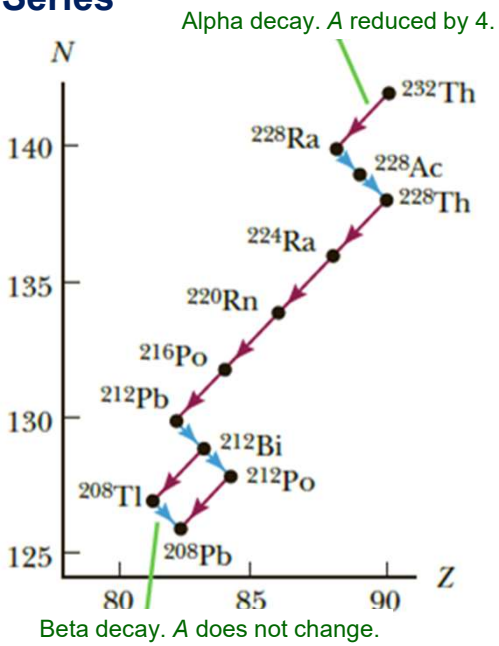
The Np-237 series is not considered as a natural series because it has decayed away over geological time due to relatively small half-life.

| Series | Starting Isotope | Half-life (years) | Stable End Product |
|---|---|---|---|
| Uranium | $^{238}_{92}U$ | $4.47 \times 10^9$ | $^{206}_{82}Pb$ |
| Actinium | $^{235}_{92}U$ | $7.04 \times 10^8$ | $^{207}_{82}Pb$ |
| Thorium | $^{232}_{90}Th$ | $1.41 \times 10^{10}$ | $^{208}_{82}Pb$ |
| Neptunium | $^{237}_{93}Np$ | $2.14 \times 10^6$ | $^{209}_{82}Pb$ |

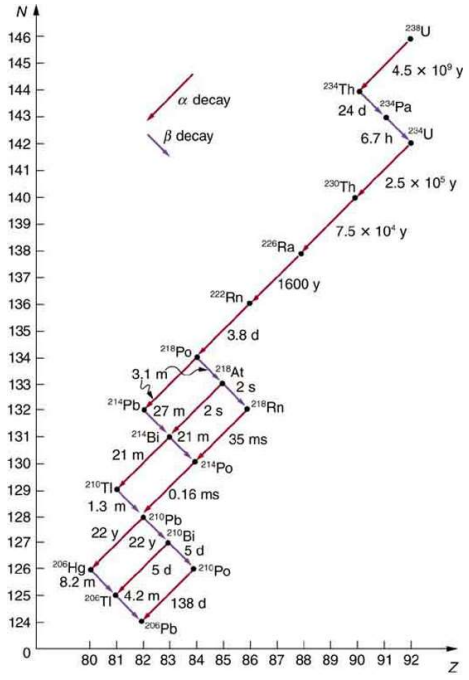





## Thorium Series



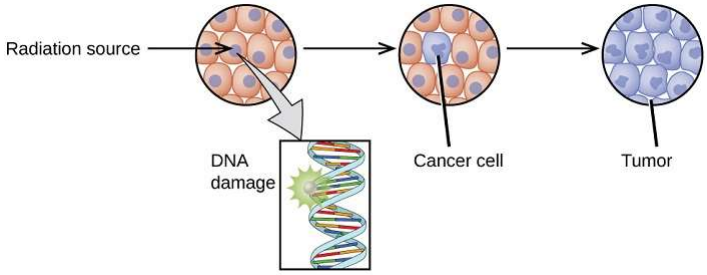
## Uranium Series

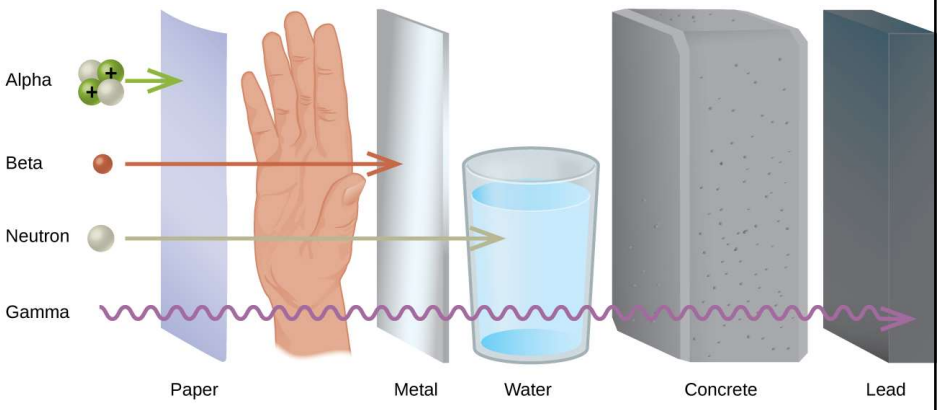



## Biological Effects of Radiation

All radioactive nuclides emit high-energy particles or electromagnetic waves. When these radiation meet living cells, they can cause heating, break chemical bonds, or ionize molecules.

The most serious biological damage is ionization of molecules in living organisms which is done by
alpha and beta particles emitted from nuclear decay reactions.






## Units of Biological Effects of Radiation

Radiation dose is a measure of the amount of exposure to radiation.

### 1. Absorbed dose

This is a measurable, physical quantity.

Absorbed dose is the amount of radiation energy absorbed per unit mass. The mass can be anything such as water, rock, air, body muscles, etc.

$$D = \frac{E}{m}$$

Absorbed dose is measured in Gy (J $kg^{-1}$)

## 2. Equivalent Dose

Equivalent dose is calculated for each radiations..

It is adjusted for the effectiveness of the type of radiation.

$$H = D \times w_R$$

where $D$ is absorbed dose and $w_R$ is radiation weighting factor. Equivalent dose is given in mSv to an organ.

| *Type of radiation* | *Weighting factor, $w_R$* |
|---|---|
| X-rays | 1 |
| gamma rays | 1 |
| beta particles | 1 |
| slow neutrons | 5 |
| fast neutrons | 10 |
| alpha particles | 20 |

## 3. Effective Dose

Effective dose is a quantity used for radiological protection. It is the addition of the equivalent doses in all relevant tissues and organs of the human body.

It is adjusted for the sensitivity of the organ to radiation.
It is a single number giving the overall risk.

Effective dose is also given in mSv.

## The tissue weighting factors

Effective dose (sievert (Sv)) = Σ (Tissue weighting factor × Equivalent dose)

$W_T$ = 0.12 (six): stomach, colon, lung, bone marrow (red), breast, and remainder tissues* (six)

$W_T$ = 0.08: gonads

$W_T$ = 0.04 (four): urinary bladder, esophagus, liver, thyroid

$W_T$ = 0.01 (our): bone surface, skin, brain, salivary glands

The sum of tissue weighting factors for all organs and tissues in the human body equals one: (0.12 x 6) + 0.08 + (0.04 x 4) + (0.01 x 4) = 1.

